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Course code:CSA0604

Course name: Data Handling and Visualization

Set 11: Product Category Analysis

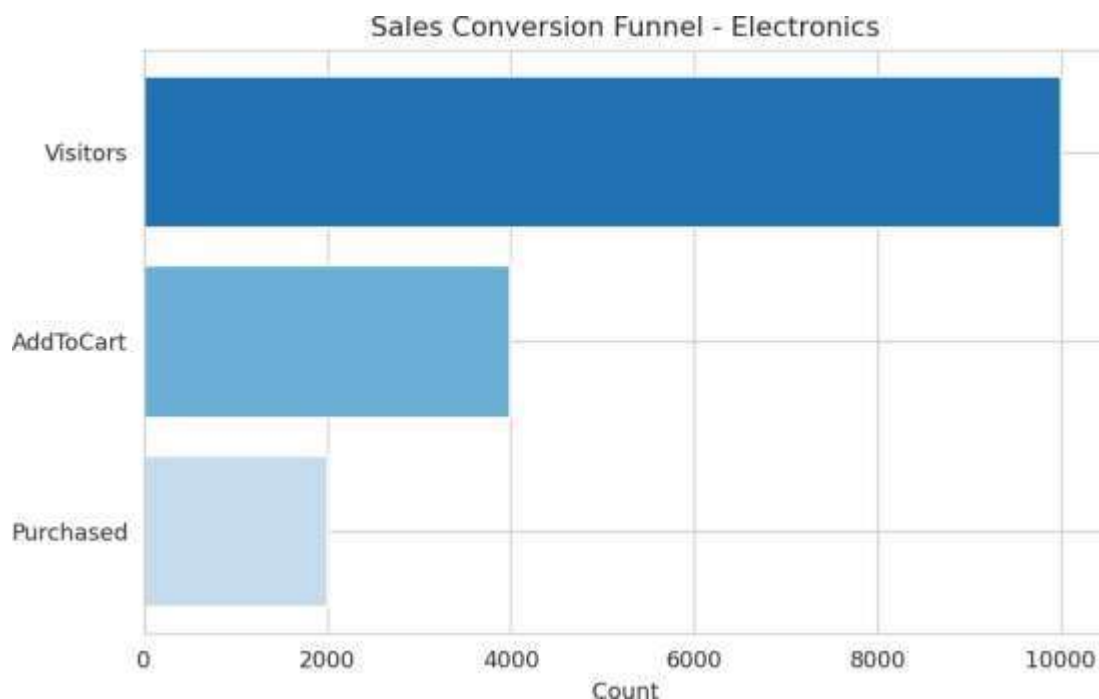
Question 1

Generate a funnel chart to analyze the sales conversion process for each product category. Label the stages and title the chart.

R Code

```
# 1. Funnel chart - sales conversion process for each category
funnel_df <- df %>% filter(Category == "Electronics")
plot_ly(
  type = "funnel",
  y = funnel_df$Stage,
  x = funnel_df$StageValue
) %>% layout(title = "Sales Conversion Funnel - Electronics")
# htmlwidgets::saveWidget as needed, e.g.:
# htmlwidgets::saveWidget(p, "funnel_chart_electronics.html")
```

Output



Question 2

Build a table to display the sales data for each product category. Label the table elements.

R Code

```
# 2. Table - sales data for each product category
cat_table <- category_sales %>% gt() %>% tab_header(title = "Sales by Product Category")
gtsave(cat_table, "table_category_sales.png")
print(category_sales)
```

Output

Sales by Product Category

Category	Sales	Stage	StageValue
Electronics	50000	Visitors	10000
Clothing	35000	Visitors	8000
Appliances	40000	Visitors	6000

Question 4

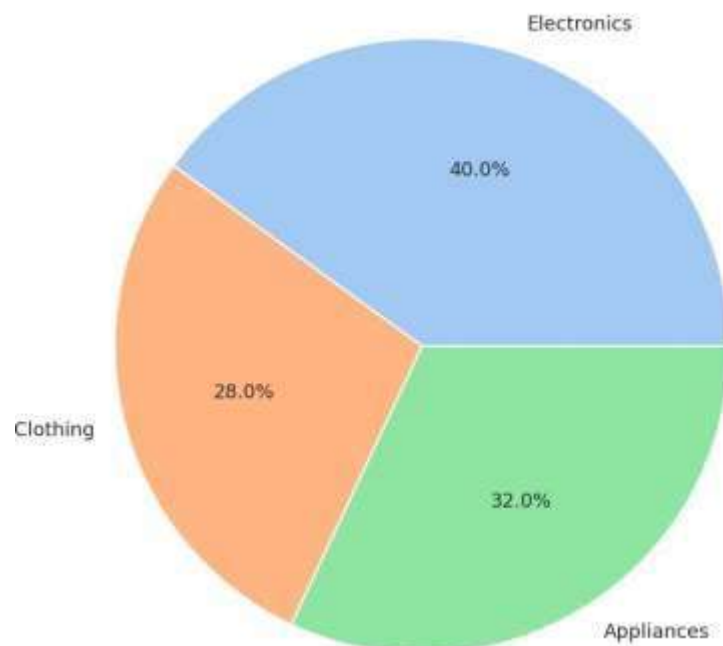
Create a pie chart to represent the distribution of sales across product categories. Include labels.

R Code

```
# 4. Pie chart - distribution of sales across categories
ggplot(category_sales, aes(x = "", y = Sales, fill = Category)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar("y") +
  labs(title = "Sales Distribution Across Product Categories") +
  theme_void()
ggsave("pie_chart_category_sales.png", width = 7, height = 5)
```

Output

Sales Distribution Across Product Categories



Set 12: Website Traffic

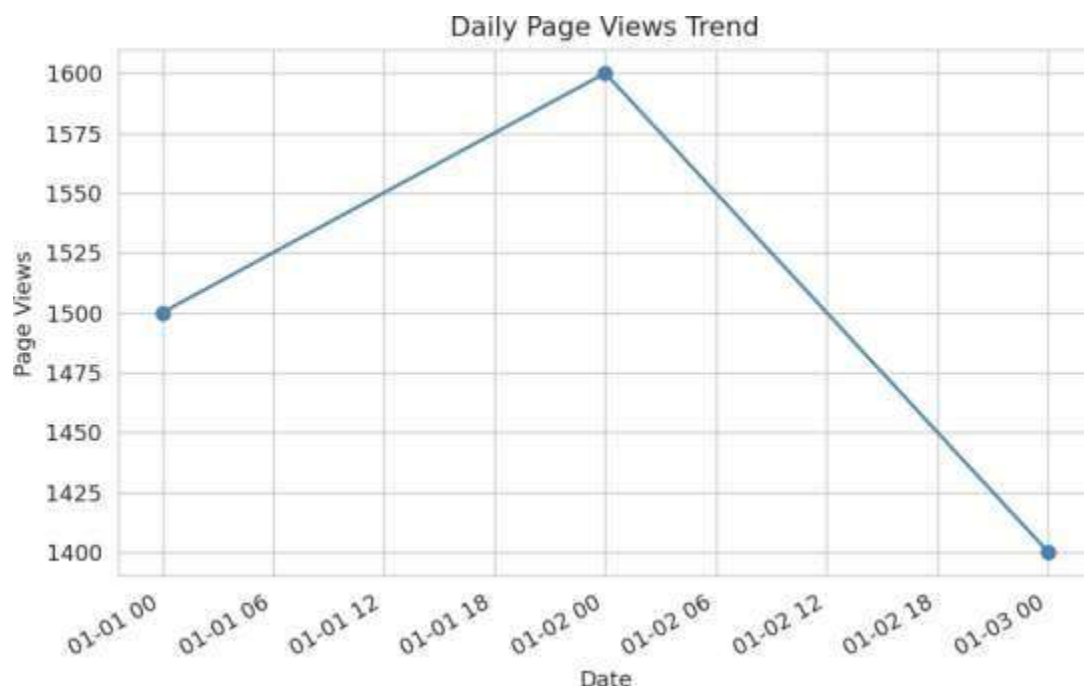
Question 1

Create a line chart to visualize the trend in daily page views over time. Label the axes and title the chart.

R Code

```
# 1. Line chart - trend in daily page views
ggplot(df, aes(x = Date, y = PageViews)) +
  geom_line(color = "steelblue", linewidth = 1) +
  geom_point() +
  labs(title = "Daily Page Views Trend", x = "Date", y = "Page Views") +
  theme_minimal()
ggsave("line_chart_page_views.png", width = 7, height = 5)
```

Output



Question 2

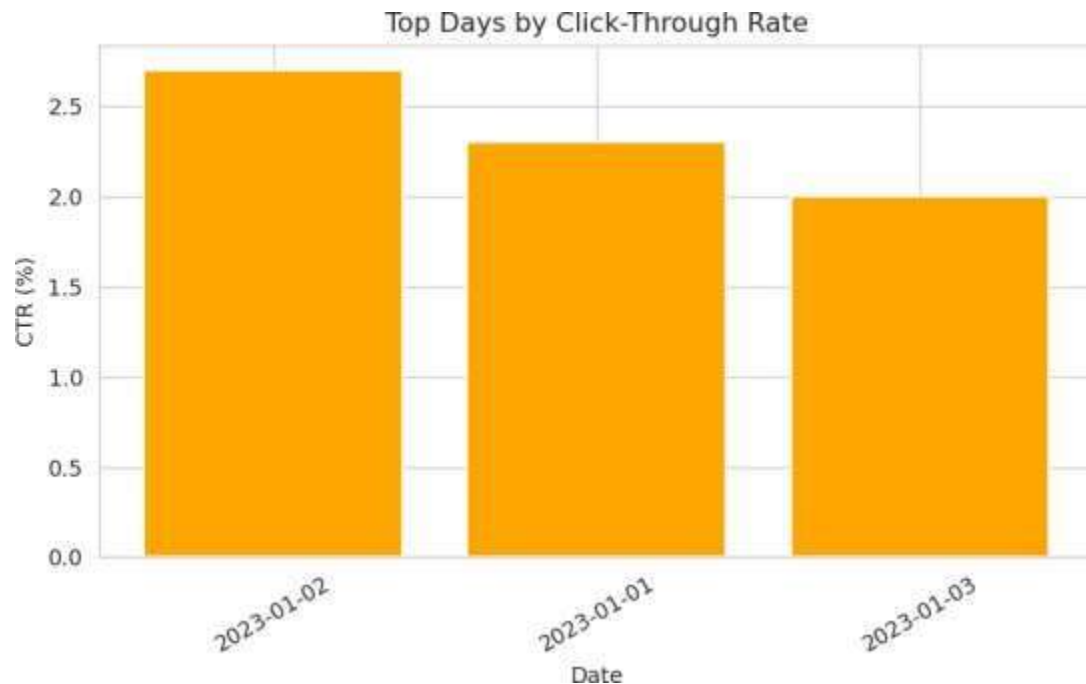
Generate a bar chart showing the top N days with the highest click-through rates. Label the chart elements.

R Code

```
# 2. Bar chart - top N days with highest click-through rates
top_n_days <- df %>% arrange(desc(ClickThroughRate))
ggplot(top_n_days, aes(x = reorder(as.character(Date), -ClickThroughRate), y =
  ClickThroughRate)) +
  geom_bar(stat = "identity", fill = "orange") +
```

```
labs(title = "Top Days by Click-Through Rate", x = "Date", y = "CTR (%)") +
theme_minimal()
ggsave("bar_chart_top_ctr_days.png", width = 7, height = 5)
```

Output



Question 3

Build a table to show the website traffic data for each day. Label the table elements.

R Code

```
# 3. Table - website traffic data for each day
traffic_table <- df %>% gt() %>% tab_header(title = "Website Traffic Data")
gtsave(traffic_table, "table_website_traffic.png")
print(df)
```

Output

Website Traffic Data

Date	PageViews	ClickThroughRate
2023-01-01	1500	2.3
2023-01-02	1600	2.7
2023-01-03	1400	2.0

Set 13: Geographic Data

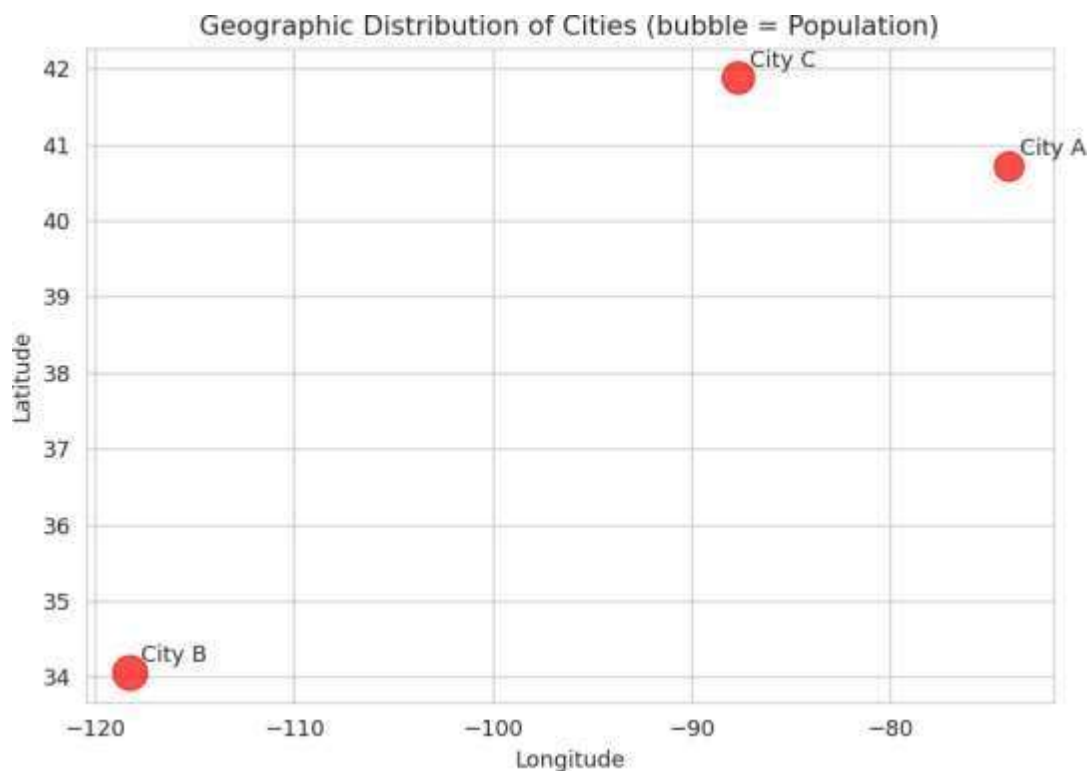
Question 1

Create a map chart to visualize the distribution of cities on a geographic map. Label the map elements.

R Code

```
# 1. Map chart - distribution of cities on a geographic map
if (requireNamespace("maps", quietly = TRUE)) {
  library(maps)
  usa_map <- map_data("state")
  ggplot() +
    geom_polygon(data = usa_map, aes(x = long, y = lat, group = group),
      fill = "gray95", color = "white") +
    geom_point(data = df, aes(x = Longitude, y = Latitude, size = Population), color =
      "red") +
    geom_text(data = df, aes(x = Longitude, y = Latitude, label = City), vjust = -1) +
    labs(title = "Geographic Distribution of Cities", x = "Longitude", y = "Latitude") +
    theme_minimal()
  ggsave("map_chart_cities.png", width = 7, height = 5)
}
```

Output



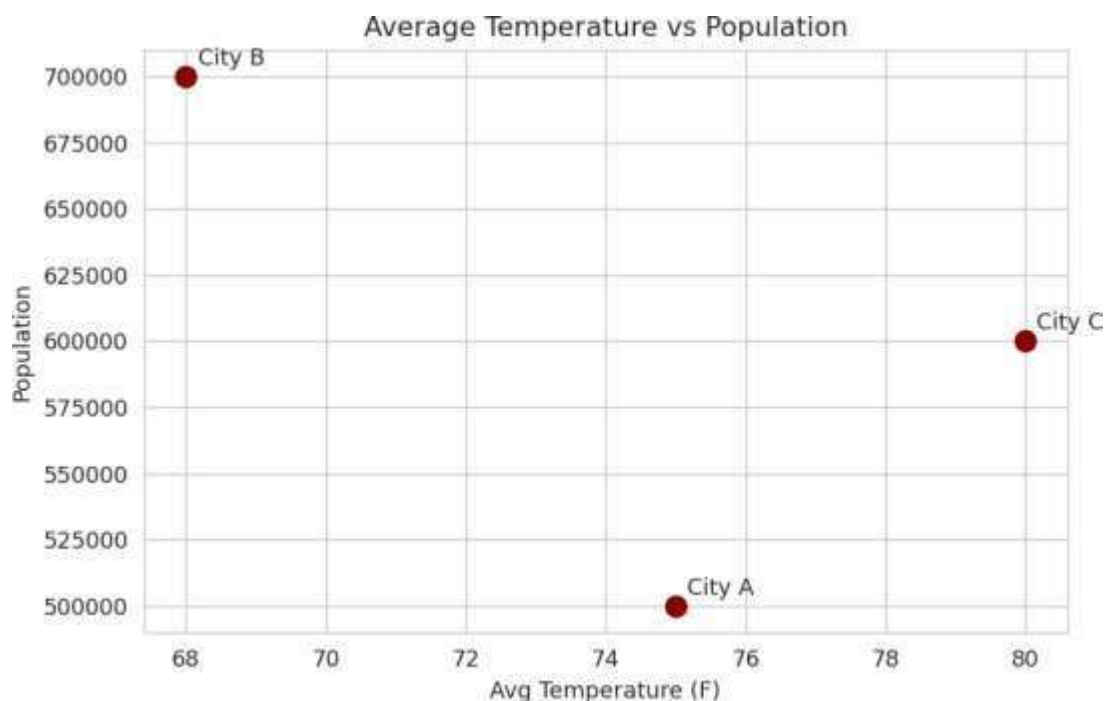
Question 2

Generate a scatter plot to explore the relationship between average temperature and population size. Explain any insights.

R Code

```
# 2. Scatter plot - average temperature vs population size
ggplot(df, aes(x = AvgTemperature, y = Population)) +
  geom_point(color = "darkred", size = 3) +
  geom_text(aes(label = City), vjust = -1) +
  labs(title = "Average Temperature vs Population", x = "Avg Temperature (F)", y =
"Population") +
  theme_minimal()
ggsave("scatter_temp_vs_population.png", width = 7, height = 5)
cat("Insight: Correlation between temperature and population =", cor(df$AvgTemperature,
df$Population), "\n")
```

Output



Question 3

Build a table to display the geographic data for each city. Label the table elements.

R Code

```
# 3. Table - geographic data for each city
geo_table <- df %>% select(City, Population, AvgTemperature, Elevation) %>%
gt() %>% tab_header(title = "Geographic Data by City")
gtsave(geo_table, "table_geographic_data.png")
print(df)
```

Output

Geographic Data by City

City	Population	AvgTemperature	Elevation
City A	500000	75	1000
City B	700000	68	800
City C	600000	80	1200

Set 14: Survey Results

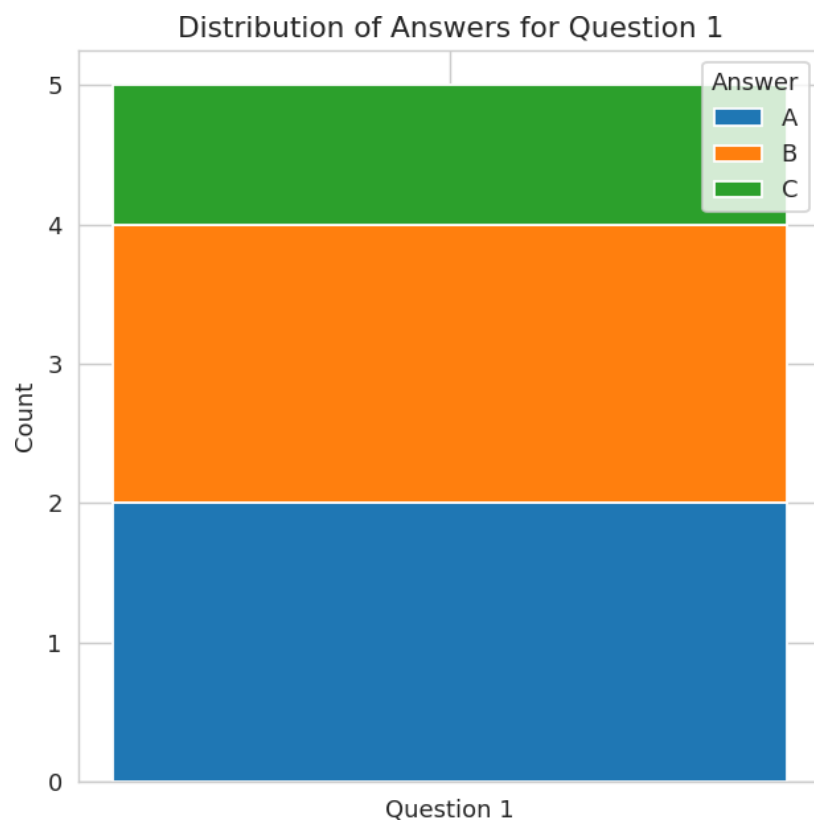
Question 1

Create a stacked bar chart to display the distribution of answers for Question 1. Label the chart elements.

R Code

```
# 1. Stacked bar chart - distribution of answers for Question 1
q1_counts <- df %>% count(Question1)
ggplot(q1_counts, aes(x = "Question1", y = n, fill = Question1)) +
  geom_bar(stat = "identity", position = "stack") +
  labs(title = "Distribution of Answers for Question 1", x = "", y = "Count") +
  theme_minimal()
ggsave("stacked_bar_question1.png", width = 7, height = 5)
```

Output



Question 2

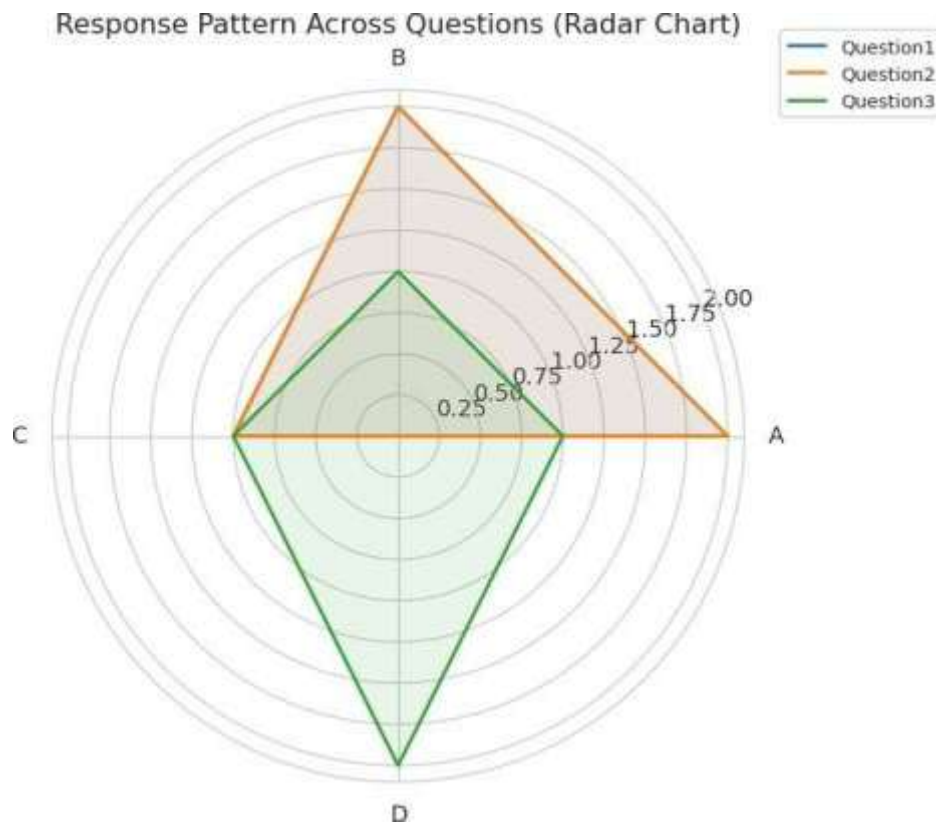
Generate a radar chart to visualize the overall pattern of responses across all three questions.

R Code

```
# 2. Radar chart - overall pattern of responses across all three questions
df_long <- df %>% pivot_longer(cols = c(Question1, Question2, Question3),
```

```
names_to = "Question", values_to = "Answer")
radar_data <- df_long %>% count(Question, Answer) %>% pivot_wider(names_from = Answer,
values_from = n, values_fill = 0)
radar_matrix <- as.data.frame(radar_data[,-1])
rownames(radar_matrix) <- radar_data$Question
radar_matrix <- rbind(apply(radar_matrix, 2, max) + 1, 0, radar_matrix)
png("radar_chart_responses.png", width = 700, height = 500)
radarchart(radar_matrix, axistype = 1, title = "Response Pattern Across Questions")
dev.off()
```

Output



Question 3

Build a table to show the survey response data for each respondent. Label the table elements.

R Code

```
# 3. Table - survey response data for each respondent
resp_table <- df %>% gt() %>% tab_header(title = "Survey Response Data")
gtsave(resp_table, "table_survey_results.png")
print(df)
```

Output

Survey Response Data

Respondent	Question1	Question2	Question3
1	A	B	C
2	B	A	D
3	C	A	B
4	A	B	D
5	B	C	A

Set 15: Product Sales Analysis

Question 1

Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.

R Code

```
# 1. Grouped bar chart - sales of each product for Q1
ggplot(df_long, aes(x = Month, y = Sales, fill = ProductName)) +
  geom_bar(stat = "identity", position = "dodge") +
  labs(title = "Q1 Sales by Product", x = "Month", y = "Sales ($)") +
  theme_minimal()
ggsave("grouped_bar_q1_sales.png", width = 7, height = 5)
```

Output



Question 2

Generate a stacked area chart to represent the overall sales trend for all products over the three months.

R Code

```
# 2. Stacked area chart - overall sales trend for all products
ggplot(df_long, aes(x = Month, y = Sales, fill = ProductName, group = ProductName)) +
  geom_area(position = "stack") +
  labs(title = "Overall Sales Trend (Stacked Area)", x = "Month", y = "Sales ($)") +
  theme_minimal()
```

```
ggsave("stacked_area_sales_trend.png", width = 7, height = 5)
```

Output



Question 3

Build a table to show the monthly sales data for each product. Label the table elements.

R Code

```
# 3. Table - monthly sales data for each product
sales_table <- df %>%
  gt() %>%
  tab_header(title = "Monthly Sales Data by Product")
gtsave(sales_table, "table_monthly_sales.png")
print(df)
```

Output

Monthly Sales Data by Product

ProductID	ProductName	JanuarySales	FebruarySales	MarchSales
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100